

Quantum circuit

1. Implement the JW mapping for the NO orbitals for ladders operators and for the Hamiltonian. (done)
2. In the ket part we will keep the determinants without convert it in the vacuum basis, this save complexity in the part of the JW mapping for the product of creation operators. (done)
3. In the bra part, we choose formulate the state in the vacuum basis for appear the annihilation operators and from that we use the NO JW mapping. (done)
4. The different determinants in the ket part will be implemented with proper X gates acting in different qubits, which will be the U_J .
5. The bra part the U_I will be containing the combination of the Pauli string and matrix elements of overlap matrix.

$$|\psi_{0,0;1}^4\rangle = \left(\hat{a}_1^\dagger \hat{a}_3^\dagger \hat{a}_6^\dagger \hat{a}_8^\dagger - \hat{a}_1^\dagger \hat{a}_4^\dagger \hat{a}_6^\dagger \hat{a}_7^\dagger - \hat{a}_2^\dagger \hat{a}_3^\dagger \hat{a}_5^\dagger \hat{a}_8^\dagger + \hat{a}_2^\dagger \hat{a}_4^\dagger \hat{a}_5^\dagger \hat{a}_7^\dagger \right) |00000000\rangle$$

$$|\psi_{0,0;1}^4\rangle = (|10100101\rangle - |10010110\rangle - |01101001\rangle + |01011010\rangle)$$

$$|\psi_{0,0;2}^4\rangle = \left(\hat{a}_1^\dagger \hat{a}_2^\dagger \hat{a}_7^\dagger \hat{a}_8^\dagger - \hat{a}_1^\dagger \hat{a}_3^\dagger \hat{a}_6^\dagger \hat{a}_8^\dagger - \hat{a}_2^\dagger \hat{a}_4^\dagger \hat{a}_5^\dagger \hat{a}_7^\dagger + \hat{a}_3^\dagger \hat{a}_4^\dagger \hat{a}_5^\dagger \hat{a}_7^\dagger \right) |00000000\rangle$$

$$|\psi_{0,0;2}^4\rangle = (|11000011\rangle - |10100101\rangle - |01011010\rangle + |00111100\rangle)$$

$$|\psi_{SCGV B}\rangle = c_1 |\psi_{0,0;1}^4\rangle + c_2 |\psi_{0,0;2}^4\rangle$$

$$\begin{aligned} (|\psi_{0,0;1}^4\rangle &= (|10100101\rangle - |10010110\rangle - |01101001\rangle + |01011010\rangle))^{\dagger} = (\langle 00000000| (a_8 a_6 a_3 a_1 - a_7 a_6 a_4 a_1 - a_8 a_5 a_3 a_2 + a_7 a_5 a_4 a_2)) \\ (|\psi_{0,0;2}^4\rangle &= (|11000011\rangle - |10100101\rangle - |01011010\rangle + |00111100\rangle))^{\dagger} = (\langle 00000000| (a_8 a_7 a_2 a_1 - a_8 a_6 a_3 a_1 - a_7 a_5 a_4 a_2 + a_6 a_5 a_4 a_3)) \end{aligned}$$

Overlap matrices

$$\langle \psi_{0,0;2}^4 | \psi_{0,0;2}^4 \rangle, \quad \langle \psi_{0,0;1}^4 | \psi_{0,0;1}^4 \rangle, \quad \langle \psi_{0,0;1}^4 | \psi_{0,0;2}^4 \rangle, \quad \langle \psi_{0,0;2}^4 | \psi_{0,0;1}^4 \rangle.$$

$$\begin{aligned} &\langle \psi_2 | \psi_2 \rangle, \langle \psi_3 | \psi_3 \rangle, \langle \psi_4 | \psi_4 \rangle, \langle \psi_5 | \psi_5 \rangle, \langle \psi_6 | \psi_6 \rangle, \langle \psi_3 | \psi_1 \rangle, \langle \psi_3 | \psi_2 \rangle, \langle \psi_4 | \psi_1 \rangle, \langle \psi_4 | \psi_2 \rangle, \langle \psi_4 | \psi_3 \rangle, \\ &\langle \psi_5 | \psi_1 \rangle, \langle \psi_5 | \psi_2 \rangle, \langle \psi_5 | \psi_3 \rangle, \langle \psi_5 | \psi_4 \rangle, \langle \psi_6 | \psi_1 \rangle, \langle \psi_6 | \psi_2 \rangle \\ &\langle \psi_6 | \psi_3 \rangle, \langle \psi_6 | \psi_4 \rangle, \langle \psi_6 | \psi_5 \rangle, \langle \psi_1 | \psi_2 \rangle \end{aligned}$$

$$\begin{aligned} \psi_1 &= |10100101\rangle \\ \psi_2 &= |10010110\rangle \\ \psi_3 &= |01101001\rangle \\ \psi_4 &= |01011010\rangle \\ \psi_5 &= |11000011\rangle \\ \psi_6 &= |00111100\rangle \end{aligned}$$

$$\begin{aligned} \langle \psi_1 | \psi_1 \rangle &= \langle 00000000 | \left(a_8 a_6 a_3 a_1 a_1^\dagger a_3^\dagger a_6^\dagger a_8^\dagger \right) | 00000000 \rangle = \langle 00000000 | (a_1 a_3 a_6 a_8) | 10100101 \rangle, \\ \langle \psi_1 | \psi_2 \rangle &= \langle 00000000 | \left(a_8 a_6 a_3 a_1 a_1^\dagger a_4^\dagger a_6^\dagger a_7^\dagger \right) | 00000000 \rangle = \langle 00000000 | (a_1 a_3 a_6 a_8) | 10010110 \rangle, \\ \langle \psi_1 | \psi_3 \rangle &= \langle 00000000 | \left(a_8 a_6 a_3 a_1 a_2^\dagger a_3^\dagger a_5^\dagger a_8^\dagger \right) | 00000000 \rangle = \langle 00000000 | (a_1 a_3 a_6 a_8) | 01101001 \rangle, \\ \langle \psi_1 | \psi_4 \rangle &= \langle 00000000 | \left(a_8 a_6 a_3 a_1 a_2^\dagger a_4^\dagger a_5^\dagger a_7^\dagger \right) | 00000000 \rangle = \langle 00000000 | (a_1 a_3 a_6 a_8) | 01011010 \rangle, \\ \langle \psi_1 | \psi_5 \rangle &= \langle 00000000 | \left(a_8 a_6 a_3 a_1 a_1^\dagger a_2^\dagger a_7^\dagger a_8^\dagger \right) | 00000000 \rangle = \langle 00000000 | (a_1 a_3 a_6 a_8) | 11000011 \rangle, \\ \langle \psi_1 | \psi_6 \rangle &= \langle 00000000 | \left(a_8 a_6 a_3 a_1 a_1^\dagger a_3^\dagger a_6^\dagger a_7^\dagger \right) | 00000000 \rangle = \langle 00000000 | (a_1 a_3 a_6 a_8) | 00111100 \rangle, \end{aligned}$$

[illegible]

JW mapping:

Below are the explicit annihilation terms for the operators $(\hat{a}_1)$ to $(\hat{a}_8)$, including the matrix elements from the overlap matrix (S_{qr}) for non-orthogonal orbitals.

Annihilation Operator for Orbital 1 ($\hat{a}_1$):

$$\begin{aligned}\hat{a}_1 = & S_{11} \frac{X_1}{2} \frac{iY_1}{2} \otimes I_2 \otimes I_3 \otimes I_4 \otimes I_5 \otimes I_6 \otimes I_7 \otimes I_8 \\ & + S_{12} Z_1 \otimes \frac{X_2 + iY_2}{2} \otimes I_3 \otimes I_4 \otimes I_5 \otimes I_6 \otimes I_7 \otimes I_8 \\ & + S_{13} Z_1 \otimes Z_2 \otimes \frac{X_3 + iY_3}{2} \otimes I_4 \otimes I_5 \otimes I_6 \otimes I_7 \otimes I_8\end{aligned}$$

Annihilation Operator for Orbital 5 ($\hat{a}_5$):

$$\begin{aligned}
\hat{a}_5 = & S_{51} \frac{X_1+iY_1}{2} \otimes I_2 \otimes I_3 \otimes I_4 \otimes I_5 \otimes I_6 \otimes I_7 \otimes I_8 \\
& + S_{52} Z_1 \otimes \frac{X_2+iY_2}{2} \otimes I_3 \otimes I_4 \otimes I_5 \otimes I_6 \otimes I_7 \otimes I_8 \\
& + S_{53} Z_1 \otimes Z_2 \otimes \frac{X_3+iY_3}{2} \otimes I_4 \otimes I_5 \otimes I_6 \otimes I_7 \otimes I_8 \\
& + S_{54} Z_1 \otimes Z_2 \otimes Z_3 \otimes \frac{X_4+iY_4}{2} \otimes I_5 \otimes I_6 \otimes I_7 \otimes I_8 \\
& + S_{55} Z_1 \otimes Z_2 \otimes Z_3 \otimes Z_4 \otimes \frac{X_5+iY_5}{2} \otimes I_6 \otimes I_7 \otimes I_8 \\
& + S_{56} Z_1 \otimes Z_2 \otimes Z_3 \otimes Z_4 \otimes Z_5 \otimes \frac{X_6+iY_6}{2} \otimes I_7 \otimes I_8 \\
& + S_{57} Z_1 \otimes Z_2 \otimes Z_3 \otimes Z_4 \otimes Z_5 \otimes Z_6 \otimes \frac{X_7+iY_7}{2} \otimes I_8 \\
& + S_{58} Z_1 \otimes Z_2 \otimes Z_3 \otimes Z_4 \otimes Z_5 \otimes Z_6 \otimes Z_7 \otimes \frac{X_8+iY_8}{2}
\end{aligned}$$

Annihilation Operator for Orbital 6 ($\hat{a}_6$):

$$\begin{aligned}
\hat{a}_6 = & S_{61} \frac{X_1+iY_1}{2} \otimes I_2 \otimes I_3 \otimes I_4 \otimes I_5 \otimes I_6 \otimes I_7 \otimes I_8 \\
& + S_{62} Z_1 \otimes \frac{X_2+iY_2}{2} \otimes I_3 \otimes I_4 \otimes I_5 \otimes I_6 \otimes I_7 \otimes I_8 \\
& + S_{63} Z_1 \otimes Z_2 \otimes \frac{X_3+iY_3}{2} \otimes I_4 \otimes I_5 \otimes I_6 \otimes I_7 \otimes I_8 \\
& + S_{64} Z_1 \otimes Z_2 \otimes Z_3 \otimes \frac{X_4+iY_4}{2} \otimes I_5 \otimes I_6 \otimes I_7 \otimes I_8 \\
& + S_{65} Z_1 \otimes Z_2 \otimes Z_3 \otimes Z_4 \otimes \frac{X_5+iY_5}{2} \otimes I_6 \otimes I_7 \otimes I_8 \\
& + S_{66} Z_1 \otimes Z_2 \otimes Z_3 \otimes Z_4 \otimes Z_5 \otimes \frac{X_6+iY_6}{2} \otimes I_7 \otimes I_8 \\
& + S_{67} Z_1 \otimes Z_2 \otimes Z_3 \otimes Z_4 \otimes Z_5 \otimes Z_6 \otimes \frac{X_7+iY_7}{2} \otimes I_8 \\
& + S_{68} Z_1 \otimes Z_2 \otimes Z_3 \otimes Z_4 \otimes Z_5 \otimes Z_6 \otimes Z_7 \otimes \frac{X_8+iY_8}{2}
\end{aligned}$$

Annihilation Operator for Orbital 7 ($\hat{a}_7$):

$$\begin{aligned}
\hat{a}_7 = & S_{71} \frac{X_1+iY_1}{2} \otimes I_2 \otimes I_3 \otimes I_4 \otimes I_5 \otimes I_6 \otimes I_7 \otimes I_8 \\
& + S_{72} Z_1 \otimes \frac{X_2+iY_2}{2} \otimes I_3 \otimes I_4 \otimes I_5 \otimes I_6 \otimes I_7 \otimes I_8 \\
& + S_{73} Z_1 \otimes Z_2 \otimes \frac{X_3+iY_3}{2} \otimes I_4 \otimes I_5 \otimes I_6 \otimes I_7 \otimes I_8 \\
& + S_{74} Z_1 \otimes Z_2 \otimes Z_3 \otimes \frac{X_4+iY_4}{2} \otimes I_5 \otimes I_6 \otimes I_7 \otimes I_8 \\
& + S_{75} Z_1 \otimes Z_2 \otimes Z_3 \otimes Z_4 \otimes \frac{X_5+iY_5}{2} \otimes I_6 \otimes I_7 \otimes I_8 \\
& + S_{76} Z_1 \otimes Z_2 \otimes Z_3 \otimes Z_4 \otimes Z_5 \otimes \frac{X_6+iY_6}{2} \otimes I_7 \otimes I_8 \\
& + S_{77} Z_1 \otimes Z_2 \otimes Z_3 \otimes Z_4 \otimes Z_5 \otimes Z_6 \otimes \frac{X_7+iY_7}{2} \otimes I_8 \\
& + S_{78} Z_1 \otimes Z_2 \otimes Z_3 \otimes Z_4 \otimes Z_5 \otimes Z_6 \otimes Z_7 \otimes \frac{X_8+iY_8}{2}
\end{aligned}$$

Annihilation Operator for Orbital 8 ($\hat{a}_8$):

$$\begin{aligned}
\hat{a}_8 = & S_{81} \frac{X_1+iY_1}{2} \otimes I_2 \otimes I_3 \otimes I_4 \otimes I_5 \otimes I_6 \otimes I_7 \otimes I_8 \\
& + S_{82} Z_1 \otimes \frac{X_2+iY_2}{2} \otimes I_3 \otimes I_4 \otimes I_5 \otimes I_6 \otimes I_7 \otimes I_8 \\
& + S_{83} Z_1 \otimes Z_2 \otimes \frac{X_3+iY_3}{2} \otimes I_4 \otimes I_5 \otimes I_6 \otimes I_7 \otimes I_8 \\
& + S_{84} Z_1 \otimes Z_2 \otimes Z_3 \otimes \frac{X_4+iY_4}{2} \otimes I_5 \otimes I_6 \otimes I_7 \otimes I_8 \\
& + S_{85} Z_1 \otimes Z_2 \otimes Z_3 \otimes Z_4 \otimes \frac{X_5+iY_5}{2} \otimes I_6 \otimes I_7 \otimes I_8 \\
& + S_{86} Z_1 \otimes Z_2 \otimes Z_3 \otimes Z_4 \otimes Z_5 \otimes \frac{X_6+iY_6}{2} \otimes I_7 \otimes I_8 \\
& + S_{87} Z_1 \otimes Z_2 \otimes Z_3 \otimes Z_4 \otimes Z_5 \otimes Z_6 \otimes \frac{X_7+iY_7}{2} \otimes I_8 \\
& + S_{88} Z_1 \otimes Z_2 \otimes Z_3 \otimes Z_4 \otimes Z_5 \otimes Z_6 \otimes Z_7 \otimes \frac{X_8+iY_8}{2}
\end{aligned}$$

Usual JW mapping:

$$\begin{aligned}
\hat{a}_1^\dagger &= \frac{X_1-iY_1}{2} \otimes I_2 \otimes I_3 \otimes I_4 \otimes I_5 \otimes I_6 \otimes I_7 \otimes I_8 \\
\hat{a}_2^\dagger &= Z_1 \otimes \frac{X_2-iY_2}{2} \otimes I_3 \otimes I_4 \otimes I_5 \otimes I_6 \otimes I_7 \otimes I_8 \\
\hat{a}_3^\dagger &= Z_1 \otimes Z_2 \otimes \frac{X_3-iY_3}{2} \otimes I_4 \otimes I_5 \otimes I_6 \otimes I_7 \otimes I_8 \\
\hat{a}_4^\dagger &= Z_1 \otimes Z_2 \otimes Z_3 \otimes \frac{X_4-iY_4}{2} \otimes I_5 \otimes I_6 \otimes I_7 \otimes I_8 \\
\hat{a}_5^\dagger &= Z_1 \otimes Z_2 \otimes Z_3 \otimes Z_4 \otimes \frac{X_5-iY_5}{2} \otimes I_6 \otimes I_7 \otimes I_8 \\
\hat{a}_6^\dagger &= Z_1 \otimes Z_2 \otimes Z_3 \otimes Z_4 \otimes Z_5 \otimes \frac{X_6-iY_6}{2} \otimes I_7 \otimes I_8 \\
\hat{a}_7^\dagger &= Z_1 \otimes Z_2 \otimes Z_3 \otimes Z_4 \otimes Z_5 \otimes Z_6 \otimes \frac{X_7-iY_7}{2} \otimes I_8 \\
\hat{a}_8^\dagger &= Z_1 \otimes Z_2 \otimes Z_3 \otimes Z_4 \otimes Z_5 \otimes Z_6 \otimes Z_7 \otimes \frac{X_8-iY_8}{2} \\
\hat{a}_1 &= \frac{X_1+iY_1}{2} \otimes I_2 \otimes I_3 \otimes I_4 \otimes I_5 \otimes I_6 \otimes I_7 \otimes I_8 \\
\hat{a}_2 &= Z_1 \otimes \frac{X_2+iY_2}{2} \otimes I_3 \otimes I_4 \otimes I_5 \otimes I_6 \otimes I_7 \otimes I_8 \\
\hat{a}_3 &= Z_1 \otimes Z_2 \otimes \frac{X_3+iY_3}{2} \otimes I_4 \otimes I_5 \otimes I_6 \otimes I_7 \otimes I_8 \\
\hat{a}_4 &= Z_1 \otimes Z_2 \otimes Z_3 \otimes \frac{X_4+Y_4}{2} \otimes I_5 \otimes I_6 \otimes I_7 \otimes I_8 \\
\hat{a}_5 &= Z_1 \otimes Z_2 \otimes Z_3 \otimes Z_4 \otimes \frac{X_5+iY_5}{2} \otimes I_6 \otimes I_7 \otimes I_8 \\
\hat{a}_6 &= Z_1 \otimes Z_2 \otimes Z_3 \otimes Z_4 \otimes Z_5 \otimes \frac{X_6+iY_6}{2} \otimes I_7 \otimes I_8 \\
\hat{a}_7 &= Z_1 \otimes Z_2 \otimes Z_3 \otimes Z_4 \otimes Z_5 \otimes Z_6 \otimes \frac{X_7+iY_7}{2} \otimes I_8 \\
\hat{a}_8 &= Z_1 \otimes Z_2 \otimes Z_3 \otimes Z_4 \otimes Z_5 \otimes Z_6 \otimes Z_7 \otimes \frac{X_8-iY_8}{2}
\end{aligned}$$

I computed the product of the operators and did the NO JW mapping and usual JW mapping using python/Qiskit. See in the code the output. I checked with the corresponding file of the overlap matrix and the expression obtained above.

$$\begin{aligned}
f_1 &= a_1^\dagger a_3^\dagger a_6^\dagger a_8^\dagger \\
f_2 &= a_1^\dagger a_4^\dagger a_6^\dagger a_7^\dagger \\
f_3 &= a_2^\dagger a_3^\dagger a_5^\dagger a_8^\dagger \\
f_4 &= a_2^\dagger a_4^\dagger a_5^\dagger a_7^\dagger \\
f_5 &= a_1^\dagger a_5^\dagger a_7^\dagger a_8^\dagger \\
f_6 &= a_3^\dagger a_4^\dagger a_5^\dagger a_6^\dagger
\end{aligned}$$

$$\begin{aligned}
w1 &= \hat{a}_8 \hat{a}_6 \hat{a}_3 \hat{a}_1 \\
w2 &= \hat{a}_7 \hat{a}_6 \hat{a}_4 \hat{a}_1 \\
w3 &= \hat{a}_8 \hat{a}_5 \hat{a}_3 \hat{a}_2 \\
w4 &= \hat{a}_7 \hat{a}_5 \hat{a}_4 \hat{a}_2 \\
w5 &= \hat{a}_8 \hat{a}_7 \hat{a}_2 \hat{a}_1 \\
w6 &= \hat{a}_6 \hat{a}_5 \hat{a}_4 \hat{a}_3
\end{aligned}$$

Trying to simplifying it for the quantum circuit notation.

$$|\phi_0\rangle = |00000000\rangle$$

$$\langle\phi_0| = \langle 00000000|$$

$$\begin{cases} |\psi_1\rangle = f_1 |\phi_0\rangle & \langle\psi_1| = w_1 \langle\phi_0| \\ |\psi_2\rangle = f_2 |\phi_0\rangle & \langle\psi_2| = w_2 \langle\phi_0| \\ |\psi_3\rangle = f_3 |\phi_0\rangle & \langle\psi_3| = w_3 \langle\phi_0| \\ |\psi_4\rangle = f_4 |\phi_0\rangle & \langle\psi_4| = w_4 \langle\phi_0| \\ |\psi_5\rangle = f_5 |\phi_0\rangle & \langle\psi_5| = w_5 \langle\phi_0| \\ |\psi_6\rangle = f_6 |\phi_0\rangle & \langle\psi_6| = w_6 \langle\phi_0| \end{cases}$$

Some conditions to consider:

$$\begin{aligned} - \langle\psi_1 | \psi_2\rangle &= \langle\psi_2 | \psi_1\rangle \\ - \langle\psi_1 | \psi_3\rangle &= \langle\psi_3 | \psi_1\rangle \\ - \langle\psi_1 | \psi_4\rangle &= \langle\psi_4 | \psi_1\rangle \\ - \langle\psi_1 | \psi_5\rangle &= \langle\psi_5 | \psi_1\rangle \\ - \langle\psi_1 | \psi_6\rangle &= \langle\psi_6 | \psi_1\rangle \end{aligned}$$

$$\begin{aligned} - \langle\psi_3 | \psi_2\rangle &= \langle\psi_2 | \psi_3\rangle \\ - \langle\psi_2 | \psi_4\rangle &= \langle\psi_4 | \psi_2\rangle \\ - \langle\psi_5 | \psi_2\rangle &= \langle\psi_2 | \psi_5\rangle \\ - \langle\psi_6 | \psi_2\rangle &= \langle\psi_2 | \psi_6\rangle \end{aligned}$$

$$\begin{aligned} - \langle\psi_3 | \psi_4\rangle &= \langle\psi_4 | \psi_3\rangle \\ - \langle\psi_5 | \psi_3\rangle &= \langle\psi_3 | \psi_5\rangle \\ - \langle\psi_6 | \psi_3\rangle &= \langle\psi_3 | \psi_6\rangle \end{aligned}$$

$$\begin{aligned} - \langle\psi_4 | \psi_5\rangle &= \langle\psi_5 | \psi_4\rangle \\ - \langle\psi_4 | \psi_6\rangle &= \langle\psi_6 | \psi_4\rangle \\ - \langle\psi_6 | \psi_5\rangle &= \langle\psi_5 | \psi_6\rangle \end{aligned}$$

In the implementation in a quantum computer, we need to add Z gate to change of phase (add a negative sign) in the end of the quantum circuit.

$$\begin{aligned} \langle\psi_1 | \psi_1\rangle &= \langle\phi_0| w_1 f_1 |\phi_0\rangle \\ \langle\psi_1 | \psi_2\rangle &= \langle\phi_0| w_1 f_2 |\phi_0\rangle \\ \langle\psi_1 | \psi_3\rangle &= \langle\phi_0| w_1 f_3 |\phi_0\rangle \\ \langle\psi_1 | \psi_4\rangle &= \langle\phi_0| w_1 f_4 |\phi_0\rangle \\ \langle\psi_1 | \psi_5\rangle &= \langle\phi_0| w_1 f_5 |\phi_0\rangle \\ \langle\psi_1 | \psi_6\rangle &= \langle\phi_0| w_1 f_6 |\phi_0\rangle \end{aligned}$$

$$\begin{aligned} \langle\psi_2 | \psi_1\rangle &= \langle\phi_0| w_2 f_1 |\phi_0\rangle \\ \langle\psi_2 | \psi_2\rangle &= \langle\phi_0| w_2 f_2 |\phi_0\rangle \\ \langle\psi_2 | \psi_3\rangle &= \langle\phi_0| w_2 f_3 |\phi_0\rangle \\ \langle\psi_2 | \psi_4\rangle &= \langle\phi_0| w_2 f_4 |\phi_0\rangle \\ \langle\psi_2 | \psi_5\rangle &= \langle\phi_0| w_2 f_5 |\phi_0\rangle \\ \langle\psi_2 | \psi_6\rangle &= \langle\phi_0| w_2 f_6 |\phi_0\rangle \end{aligned}$$

$$\begin{aligned} \langle\psi_3 | \psi_1\rangle &= \langle\phi_0| w_3 f_1 |\phi_0\rangle \\ \langle\psi_3 | \psi_2\rangle &= \langle\phi_0| w_3 f_2 |\phi_0\rangle \\ \langle\psi_3 | \psi_3\rangle &= \langle\phi_0| w_3 f_3 |\phi_0\rangle \\ \langle\psi_3 | \psi_4\rangle &= \langle\phi_0| w_3 f_4 |\phi_0\rangle \\ \langle\psi_3 | \psi_5\rangle &= \langle\phi_0| w_3 f_5 |\phi_0\rangle \end{aligned}$$

$$\begin{aligned}
\langle \psi_6 | \psi_2 \rangle &= \langle \phi_0 | w_6 f_2 | \phi_0 \rangle = -0.1166190534E - 02 \\
\langle \psi_6 | \psi_3 \rangle &= \langle \phi_0 | w_6 f_3 | \phi_0 \rangle = -0.1166190534E - 02 \\
\langle \psi_6 | \psi_4 \rangle &= \langle \phi_0 | w_6 f_4 | \phi_0 \rangle = -0.1227232103E + 00 \\
\langle \psi_6 | \psi_5 \rangle &= \langle \phi_0 | w_6 f_5 | \phi_0 \rangle = 0.4616063641E - 1 \\
\langle \psi_6 | \psi_6 \rangle &= \langle \phi_0 | w_6 f_6 | \phi_0 \rangle = 0.3201019355E + 00
\end{aligned}$$

Results which I obtained with the Quantum Computing file “untitled 2.py”.

$$\begin{aligned}
\langle \psi_1 | \psi_1 \rangle &= \langle \phi_0 | w_1 f_1 | \phi_0 \rangle = 0.609376605 \\
\langle \psi_1 | \psi_2 \rangle &= \langle \phi_0 | w_1 f_2 | \phi_0 \rangle = 0.12272321 \\
\langle \psi_1 | \psi_3 \rangle &= \langle \phi_0 | w_1 f_3 | \phi_0 \rangle = 0.12272321 \\
\langle \psi_1 | \psi_4 \rangle &= \langle \phi_0 | w_1 f_4 | \phi_0 \rangle = 0 \\
\langle \psi_1 | \psi_5 \rangle &= \langle \phi_0 | w_1 f_5 | \phi_0 \rangle = 0.12272321 \\
\langle \psi_1 | \psi_6 \rangle &= \langle \phi_0 | w_1 f_6 | \phi_0 \rangle = 0.12272321
\end{aligned}$$

$$\begin{aligned}
\langle \psi_2 | \psi_1 \rangle &= \langle \phi_0 | w_2 f_1 | \phi_0 \rangle = 0.12272321 \\
\langle \psi_2 | \psi_2 \rangle &= \langle \phi_0 | w_2 f_2 | \phi_0 \rangle = 0.320101931 \\
\langle \psi_2 | \psi_3 \rangle &= \langle \phi_0 | w_2 f_3 | \phi_0 \rangle = 0.046160637 \\
\langle \psi_2 | \psi_4 \rangle &= \langle \phi_0 | w_2 f_4 | \phi_0 \rangle = 0.12272321 \\
\langle \psi_2 | \psi_5 \rangle &= \langle \phi_0 | w_2 f_5 | \phi_0 \rangle = -0.00116619 \\
\langle \psi_2 | \psi_6 \rangle &= \langle \phi_0 | w_2 f_6 | \phi_0 \rangle = -0.00116619
\end{aligned}$$

$$\begin{aligned}
\langle \psi_3 | \psi_1 \rangle &= \langle \phi_0 | w_3 f_1 | \phi_0 \rangle = 0.12272321 \\
\langle \psi_3 | \psi_2 \rangle &= \langle \phi_0 | w_3 f_2 | \phi_0 \rangle = 0.046160637 \\
\langle \psi_3 | \psi_3 \rangle &= \langle \phi_0 | w_3 f_3 | \phi_0 \rangle = 0.320101931 \\
\langle \psi_3 | \psi_4 \rangle &= \langle \phi_0 | w_3 f_4 | \phi_0 \rangle = 0.12272321 \\
\langle \psi_3 | \psi_5 \rangle &= \langle \phi_0 | w_3 f_5 | \phi_0 \rangle = -0.00116619 \\
\langle \psi_3 | \psi_6 \rangle &= \langle \phi_0 | w_3 f_6 | \phi_0 \rangle = -0.00116619
\end{aligned}$$

$$\begin{aligned}
\langle \psi_4 | \psi_1 \rangle &= \langle \phi_0 | w_4 f_1 | \phi_0 \rangle = 0 \\
\langle \psi_4 | \psi_2 \rangle &= \langle \phi_0 | w_4 f_2 | \phi_0 \rangle = 0.12272321 \\
\langle \psi_4 | \psi_3 \rangle &= \langle \phi_0 | w_4 f_3 | \phi_0 \rangle = 0.12272321 \\
\langle \psi_4 | \psi_4 \rangle &= \langle \phi_0 | w_4 f_4 | \phi_0 \rangle = 0.609376605 \\
\langle \psi_4 | \psi_5 \rangle &= \langle \phi_0 | w_4 f_5 | \phi_0 \rangle = 0.12272321 \\
\langle \psi_4 | \psi_6 \rangle &= \langle \phi_0 | w_4 f_6 | \phi_0 \rangle = 0.12272321
\end{aligned}$$

$$\begin{aligned}
\langle \psi_5 | \psi_1 \rangle &= \langle \phi_0 | w_5 f_1 | \phi_0 \rangle = 0.12272321 \\
\langle \psi_5 | \psi_2 \rangle &= \langle \phi_0 | w_5 f_2 | \phi_0 \rangle = -0.00116619 \\
\langle \psi_5 | \psi_3 \rangle &= \langle \phi_0 | w_5 f_3 | \phi_0 \rangle = -0.00116619 \\
\langle \psi_5 | \psi_4 \rangle &= \langle \phi_0 | w_5 f_4 | \phi_0 \rangle = 0.12272321 \\
\langle \psi_5 | \psi_5 \rangle &= \langle \phi_0 | w_5 f_5 | \phi_0 \rangle = 0.320101931 \\
\langle \psi_5 | \psi_6 \rangle &= \langle \phi_0 | w_5 f_6 | \phi_0 \rangle = 0.046160637
\end{aligned}$$

$$\begin{aligned}
\langle \psi_6 | \psi_1 \rangle &= \langle \phi_0 | w_6 f_1 | \phi_0 \rangle = 0.12272321 \\
\langle \psi_6 | \psi_2 \rangle &= \langle \phi_0 | w_6 f_2 | \phi_0 \rangle = -0.00116619 \\
\langle \psi_6 | \psi_3 \rangle &= \langle \phi_0 | w_6 f_3 | \phi_0 \rangle = -0.00116619 \\
\langle \psi_6 | \psi_4 \rangle &= \langle \phi_0 | w_6 f_4 | \phi_0 \rangle = 0.12272321 \\
\langle \psi_6 | \psi_5 \rangle &= \langle \phi_0 | w_6 f_5 | \phi_0 \rangle = 0.046160637 \\
\langle \psi_6 | \psi_6 \rangle &= \langle \phi_0 | w_6 f_6 | \phi_0 \rangle = 0.320101931
\end{aligned}$$

$$\begin{aligned}
O_{IJ} &= \langle \psi_I | \psi_J \rangle \\
\langle \psi_I | &= w_i \langle \phi_0 |, i=1 \text{ to } 6. \\
| \psi_J \rangle &= f_j | \phi_0 \rangle, j=1 \text{ to } 6.
\end{aligned}$$

Overlap matrix

$$S_{\langle\psi_1|\psi_1\rangle} = \begin{pmatrix} S_{11} & S_{13} & S_{16} & S_{18} \\ S_{31} & S_{33} & S_{36} & S_{38} \\ S_{61} & S_{623} & S_{66} & S_{68} \\ S_{81} & S_{83} & S_{86} & S_{88} \end{pmatrix} =$$

$$= S_{11}S_{33}S_{66}S_{88} - S_{11}S_{33}S_{86}^2 - S_{11}S_{63}^2S_{88} + 2S_{11}S_{63}S_{83}S_{86} - S_{11}S_{66}S_{83}^2 -$$

$$= S_{31}^2S_{66}S_{88} + S_{31}^2S_{86}^2 + 2S_{31}S_{61}S_{63}S_{88} - 2S_{31}S_{61}S_{83}S_{86} - 2S_{31}S_{63}S_{81}S_{86}$$

$$+ 2S_{31}S_{66}S_{81}S_{83} - S_{33}S_{61}^2S_{88} + 2S_{33}S_{61}S_{81}S_{86} - S_{33}S_{66}S_{81}^2 + S_{61}^2S_{83}^2 -$$

$$2S_{61}S_{63}S_{81}S_{83} + S_{63}^2S_{81}^2.$$

$$S_{\langle\psi_1|\psi_2\rangle} = \begin{pmatrix} S_{11} & S_{14} & S_{16} & S_{17} \\ S_{31} & S_{34} & S_{36} & S_{37} \\ S_{63} & S_{64} & S_{66} & S_{67} \\ S_{83} & S_{84} & S_{86} & S_{87} \end{pmatrix} =$$

$$S_{11}S_{43}S_{66}S_{87} - S_{11}S_{43}S_{76}S_{86} - S_{11}S_{63}S_{64}S_{87} + S_{11}S_{63}S_{76}S_{84} + S_{11}S_{64}S_{73}S_{86}$$

$$- S_{11}S_{66}S_{73}S_{84} - S_{31}S_{41}S_{66}S_{87} + S_{31}S_{41}S_{76}S_{86} + S_{31}S_{61}S_{64}S_{87} - S_{31}S_{61}S_{76}S_{84}$$

$$= -S_{31}S_{64}S_{71}S_{86} + S_{31}S_{66}S_{71}S_{84} + S_{41}S_{61}S_{63}S_{87} - S_{41}S_{61}S_{73}S_{86} -$$

$$S_{41}S_{63}S_{76}S_{81} + S_{41}S_{66}S_{73}S_{81} - S_{43}S_{61}^2S_{87} + S_{43}S_{61}S_{71}S_{86} + S_{43}S_{61}S_{76}S_{81} -$$

$$S_{43}S_{66}S_{71}S_{81} + S_{61}^2S_{73}S_{84} - S_{61}S_{63}S_{71}S_{84} - S_{61}S_{64}S_{73}S_{81} + S_{63}S_{64}S_{71}S_{81}.$$

$$S_{\langle\psi_1|\psi_3\rangle} = \begin{pmatrix} S_{12} & S_{13} & S_{15} & S_{18} \\ S_{32} & S_{33} & S_{35} & S_{38} \\ S_{62} & S_{63} & S_{65} & S_{68} \\ S_{82} & S_{83} & S_{85} & S_{88} \end{pmatrix} =$$

$$S_{21}S_{33}S_{65}S_{88} - S_{21}S_{33}S_{85}S_{86} - S_{21}S_{53}S_{63}S_{88} + S_{21}S_{53}S_{83}S_{86} + S_{21}S_{63}S_{83}S_{85}$$

$$- S_{21}S_{65}S_{83}^2 - S_{31}S_{32}S_{65}S_{88} + S_{31}S_{32}S_{85}S_{86} + S_{31}S_{53}S_{62}S_{88} - S_{31}S_{53}S_{82}S_{86}$$

$$- S_{31}S_{62}S_{83}S_{85} + S_{31}S_{65}S_{82}S_{83} + S_{32}S_{51}S_{63}S_{88} - S_{32}S_{51}S_{83}S_{86} -$$

$$S_{32}S_{63}S_{81}S_{85} + S_{32}S_{65}S_{81}S_{83} - S_{33}S_{51}S_{62}S_{88} + S_{33}S_{51}S_{82}S_{86} + S_{33}S_{62}S_{81}S_{85} -$$

$$S_{33}S_{65}S_{81}S_{82} + S_{51}S_{62}S_{83}^2 - S_{51}S_{63}S_{82}S_{83} - S_{53}S_{62}S_{81}S_{83} + S_{53}S_{63}S_{81}S_{82}.$$

$$S_{\langle\psi_1|\psi_4\rangle} = \begin{pmatrix} S_{11} & S_{13} & S_{16} & S_{18} \\ S_{31} & S_{33} & S_{36} & S_{38} \\ S_{61} & S_{63} & S_{66} & S_{68} \\ S_{81} & S_{83} & S_{86} & S_{88} \end{pmatrix} =$$

$$S_{21}S_{43}S_{65}S_{87} - S_{21}S_{43}S_{76}S_{85} - S_{21}S_{53}S_{64}S_{87} + S_{21}S_{53}S_{76}S_{84} + S_{21}S_{64}S_{73}S_{85} - S_{21}S_{65}S_{73}S_{84} -$$

$$S_{32}S_{41}S_{65}S_{87} + S_{32}S_{41}S_{76}S_{85} + S_{32}S_{51}S_{64}S_{87} - S_{32}S_{51}S_{76}S_{84} - S_{32}S_{64}S_{71}S_{85} + S_{32}S_{65}S_{71}S_{84} +$$

$$+ S_{41}S_{53}S_{62}S_{87} - S_{41}S_{53}S_{76}S_{82} - S_{41}S_{62}S_{73}S_{85} + S_{41}S_{65}S_{73}S_{82} - S_{43}S_{51}S_{62}S_{87} + S_{43}S_{51}S_{76}S_{82}$$

$$+ S_{43}S_{62}S_{71}S_{85} - S_{43}S_{65}S_{71}S_{82} + S_{51}S_{62}S_{73}S_{84} - S_{51}S_{64}S_{73}S_{82} - S_{53}S_{62}S_{71}S_{84} + S_{53}S_{64}S_{71}S_{82}$$

$$S_{\langle\psi_1|\psi_5\rangle} = \begin{pmatrix} S_{11} & S_{12} & S_{17} & S_{18} \\ S_{31} & S_{32} & S_{37} & S_{38} \\ S_{61} & S_{62} & S_{67} & S_{68} \\ S_{81} & S_{82} & S_{87} & S_{88} \end{pmatrix} =$$

$$S_{11}S_{32}S_{76}S_{88} - S_{11}S_{32}S_{86}S_{87} - S_{11}S_{62}S_{73}S_{88} + S_{11}S_{62}S_{83}S_{87} + S_{11}S_{73}S_{82}S_{86} - S_{11}S_{76}S_{82}S_{83}$$

$$- S_{21}S_{31}S_{76}S_{88} + S_{21}S_{31}S_{86}S_{87} + S_{21}S_{61}S_{73}S_{88} - S_{21}S_{61}S_{83}S_{87} - S_{21}S_{73}S_{81}S_{86} +$$

$$S_{21}S_{76}S_{81}S_{83} + S_{31}S_{62}S_{71}S_{88} - S_{31}S_{62}S_{81}S_{87} - S_{31}S_{71}S_{82}S_{86} + S_{31}S_{76}S_{81}S_{82} - S_{32}S_{61}S_{71}S_{88} +$$

$$S_{32}S_{61}S_{81}S_{87} + S_{32}S_{71}S_{81}S_{86} - S_{32}S_{76}S_{81}^2 + S_{61}S_{71}S_{82}S_{83} - S_{61}S_{73}S_{81}S_{82} - S_{62}S_{71}S_{81}S_{83} + S_{62}S_{73}S_{81}^2$$

$$S_{\langle\psi_1|\psi_6\rangle} = \begin{pmatrix} S_{13} & S_{14} & S_{15} & S_{16} \\ S_{33} & S_{34} & S_{35} & S_{36} \\ S_{63} & S_{64} & S_{65} & S_{66} \\ S_{83} & S_{84} & S_{85} & S_{86} \end{pmatrix} =$$

$$S_{31}S_{43}S_{65}S_{86} - S_{31}S_{43}S_{66}S_{85} - S_{31}S_{53}S_{64}S_{86} + S_{31}S_{53}S_{66}S_{84} + S_{31}S_{63}S_{64}S_{85} - S_{31}S_{63}S_{65}S_{84}$$

$$- S_{33}S_{41}S_{65}S_{86} + S_{33}S_{41}S_{66}S_{85} + S_{33}S_{51}S_{64}S_{86} - S_{33}S_{51}S_{66}S_{84} - S_{33}S_{61}S_{64}S_{85} + S_{33}S_{61}S_{65}S_{84}$$

$$+ S_{41}S_{53}S_{63}S_{86} - S_{41}S_{53}S_{66}S_{83} - S_{41}S_{63}^2S_{85} + S_{41}S_{63}S_{65}S_{83} - S_{43}S_{51}S_{63}S_{86} + S_{43}S_{51}S_{66}S_{83} +$$

$$S_{43}S_{61}S_{63}S_{85} - S_{43}S_{61}S_{65}S_{83} + S_{51}S_{63}^2S_{84} - S_{51}S_{63}S_{64}S_{83} - S_{53}S_{61}S_{63}S_{84} + S_{53}S_{61}S_{64}S_{83}$$

$$S_{\langle\psi_2|\psi_1\rangle} = \begin{pmatrix} S_{11} & S_{13} & S_{16} & S_{18} \\ S_{41} & S_{43} & S_{46} & S_{48} \\ S_{61} & S_{63} & S_{66} & S_{68} \\ S_{71} & S_{73} & S_{76} & S_{78} \end{pmatrix} =$$

$$S_{11}S_{43}S_{66}S_{87} - S_{11}S_{43}S_{76}S_{86} - S_{11}S_{63}S_{64}S_{87} + S_{11}S_{63}S_{76}S_{84} + S_{11}S_{64}S_{73}S_{86} - S_{11}S_{66}S_{73}S_{84}$$

$$- S_{31}S_{41}S_{66}S_{87} + S_{31}S_{41}S_{76}S_{86} + S_{31}S_{61}S_{64}S_{87} - S_{31}S_{61}S_{76}S_{84} - S_{31}S_{64}S_{71}S_{86} +$$

$$S_{31}S_{66}S_{71}S_{84} + S_{41}S_{61}S_{63}S_{87} - S_{41}S_{61}S_{73}S_{86} - S_{41}S_{63}S_{76}S_{81} + S_{41}S_{66}S_{73}S_{81} - S_{43}S_{61}^2S_{87} +$$

$$S_{43}S_{61}S_{71}S_{86} + S_{43}S_{61}S_{76}S_{81} - S_{43}S_{66}S_{71}S_{81} + S_{61}^2S_{73}S_{84} - S_{61}S_{63}S_{71}S_{84} - S_{61}S_{64}S_{73}S_{81} + S_{63}S_{64}S_{71}S_{81}$$

$$S_{\langle\psi_2|\psi_2\rangle} = \begin{pmatrix} S_{11} & S_{14} & S_{16} & S_{17} \\ S_{41} & S_{44} & S_{46} & S_{47} \\ S_{61} & S_{64} & S_{66} & S_{67} \\ S_{71} & S_{74} & S_{76} & S_{77} \end{pmatrix} =$$

$$S_{11}S_{44}S_{66}S_{77} - S_{11}S_{44}S_{76}^2 - S_{11}S_{64}^2S_{77} + 2S_{11}S_{64}S_{74}S_{76} - S_{11}S_{66}S_{74}^2 -$$

$$S_{41}^2S_{66}S_{77} + S_{41}^2S_{76}^2 + 2S_{41}S_{61}S_{64}S_{77} - 2S_{41}S_{61}S_{74}S_{76} - 2S_{41}S_{64}S_{71}S_{76} +$$

$$2S_{41}S_{66}S_{71}S_{74} - S_{44}S_{61}^2S_{77} + 2S_{44}S_{61}S_{71}S_{76} - S_{44}S_{66}S_{71}^2 + S_{61}^2S_{74}^2 - 2S_{61}S_{64}S_{71}S_{74} + S_{64}^2S_{71}^2$$

$$S_{\langle\psi_2|\psi_3\rangle} = \begin{pmatrix} S_{12} & S_{13} & S_{15} & S_{18} \\ S_{42} & S_{43} & S_{45} & S_{48} \\ S_{62} & S_{63} & S_{65} & S_{68} \\ S_{72} & S_{73} & S_{75} & S_{78} \end{pmatrix} =$$

$$S_{21}S_{43}S_{65}S_{87} - S_{21}S_{43}S_{75}S_{86} - S_{21}S_{54}S_{63}S_{87} + S_{21}S_{54}S_{73}S_{86} + S_{21}S_{63}S_{75}S_{84} - S_{21}S_{65}S_{73}S_{84} - S_{31}S_{42}S_{65}S_{87}$$

$$+ S_{31}S_{42}S_{75}S_{86} + S_{31}S_{54}S_{62}S_{87} - S_{31}S_{54}S_{72}S_{86} - S_{31}S_{62}S_{75}S_{84} + S_{31}S_{65}S_{72}S_{84}$$

$$+ S_{42}S_{51}S_{63}S_{87} - S_{42}S_{51}S_{73}S_{86} - S_{42}S_{63}S_{75}S_{81} + S_{43}S_{51}S_{62}S_{87} + S_{43}S_{51}S_{72}S_{86} +$$

$$S_{43}S_{62}S_{75}S_{81} - S_{43}S_{65}S_{72}S_{81} + S_{51}S_{62}S_{73}S_{84} - S_{51}S_{63}S_{72}S_{84} - S_{54}S_{62}S_{73}S_{81} + S_{54}S_{63}S_{72}S_{81}$$

$$S_{\langle\psi_2|\psi_4\rangle} = \begin{pmatrix} S_{12} & S_{14} & S_{15} & S_{17} \\ S_{42} & S_{44} & S_{45} & S_{47} \\ S_{62} & S_{64} & S_{65} & S_{67} \\ S_{72} & S_{74} & S_{75} & S_{77} \end{pmatrix} =$$

$$S_{\langle\psi_2|\psi_5\rangle} = \begin{pmatrix} S_{11} & S_{12} & S_{17} & S_{18} \\ S_{41} & S_{42} & S_{47} & S_{48} \\ S_{61} & S_{62} & S_{67} & S_{68} \\ S_{71} & S_{72} & S_{77} & S_{78} \end{pmatrix} =$$

$$S_{\langle\psi_2|\psi_6\rangle} = \begin{pmatrix} S_{13} & S_{14} & S_{15} & S_{16} \\ S_{43} & S_{44} & S_{45} & S_{46} \\ S_{63} & S_{64} & S_{65} & S_{66} \\ S_{73} & S_{74} & S_{75} & S_{76} \end{pmatrix} =$$

$$S_{\langle\psi_3|\psi_1\rangle} = \begin{pmatrix} S_{21} & S_{23} & S_{26} & S_{28} \\ S_{31} & S_{33} & S_{36} & S_{38} \\ S_{51} & S_{53} & S_{56} & S_{58} \\ S_{81} & S_{83} & S_{86} & S_{88} \end{pmatrix} =$$

$$S_{\langle\psi_3|\psi_2\rangle} = \begin{pmatrix} S_{21} & S_{24} & S_{26} & S_{27} \\ S_{31} & S_{34} & S_{36} & S_{37} \\ S_{51} & S_{54} & S_{56} & S_{57} \\ S_{81} & S_{84} & S_{86} & S_{87} \end{pmatrix} =$$

$$S_{\langle\psi_3|\psi_4\rangle} = \begin{pmatrix} S_{22} & S_{24} & S_{25} & S_{27} \\ S_{32} & S_{34} & S_{35} & S_{37} \\ S_{52} & S_{54} & S_{55} & S_{57} \\ S_{82} & S_{84} & S_{85} & S_{87} \end{pmatrix} =$$

$$S_{\langle\psi_3|\psi_5\rangle} = \begin{pmatrix} S_{21} & S_{22} & S_{27} & S_{28} \\ S_{31} & S_{32} & S_{37} & S_{38} \\ S_{51} & S_{52} & S_{57} & S_{58} \\ S_{81} & S_{82} & S_{87} & S_{88} \end{pmatrix} =$$

$$S_{\langle\psi_3|\psi_6\rangle} = \begin{pmatrix} S_{23} & S_{24} & S_{25} & S_{26} \\ S_{33} & S_{34} & S_{35} & S_{36} \\ S_{53} & S_{54} & S_{55} & S_{56} \\ S_{83} & S_{84} & S_{85} & S_{86} \end{pmatrix} =$$

$$S_{\langle \psi_4 | \psi_1 \rangle} = \begin{pmatrix} S_{21} & S_{23} & S_{26} & S_{28} \\ S_{41} & S_{43} & S_{46} & S_{48} \\ S_{51} & S_{53} & S_{56} & S_{58} \\ S_{71} & S_{73} & S_{76} & S_{78} \end{pmatrix} =$$

$$S_{\langle \psi_4 | \psi_2 \rangle} = \begin{pmatrix} S_{21} & S_{24} & S_{26} & S_{27} \\ S_{41} & S_{44} & S_{46} & S_{47} \\ S_{51} & S_{54} & S_{56} & S_{57} \\ S_{71} & S_{74} & S_{76} & S_{77} \end{pmatrix} =$$

$$S_{\langle \psi_4 | \psi_3 \rangle} = \begin{pmatrix} S_{22} & S_{23} & S_{25} & S_{28} \\ S_{42} & S_{43} & S_{45} & S_{48} \\ S_{52} & S_{53} & S_{55} & S_{58} \\ S_{72} & S_{73} & S_{75} & S_{78} \end{pmatrix} =$$

$$S_{\langle \psi_4 | \psi_4 \rangle} = \begin{pmatrix} S_{22} & S_{24} & S_{25} & S_{27} \\ S_{42} & S_{44} & S_{45} & S_{47} \\ S_{52} & S_{54} & S_{55} & S_{57} \\ S_{72} & S_{74} & S_{75} & S_{77} \end{pmatrix} =$$

$$S_{\langle \psi_4 | \psi_5 \rangle} = \begin{pmatrix} S_{21} & S_{22} & S_{27} & S_{28} \\ S_{41} & S_{42} & S_{47} & S_{48} \\ S_{51} & S_{52} & S_{57} & S_{58} \\ S_{71} & S_{72} & S_{77} & S_{78} \end{pmatrix} =$$

$$S_{\langle \psi_4 | \psi_6 \rangle} = \begin{pmatrix} S_{23} & S_{24} & S_{25} & S_{26} \\ S_{43} & S_{44} & S_{45} & S_{46} \\ S_{53} & S_{54} & S_{55} & S_{56} \\ S_{73} & S_{74} & S_{75} & S_{76} \end{pmatrix} =$$

$$S_{\langle \psi_5 | \psi_1 \rangle} = \begin{pmatrix} S_{11} & S_{13} & S_{16} & S_{18} \\ S_{21} & S_{23} & S_{26} & S_{28} \\ S_{71} & S_{73} & S_{76} & S_{78} \\ S_{81} & S_{83} & S_{86} & S_{88} \end{pmatrix} =$$

$$S_{\langle \psi_5 | \psi_2 \rangle} = \begin{pmatrix} S_{11} & S_{14} & S_{16} & S_{17} \\ S_{21} & S_{24} & S_{26} & S_{27} \\ S_{71} & S_{74} & S_{76} & S_{77} \\ S_{81} & S_{84} & S_{86} & S_{87} \end{pmatrix} =$$

$$S_{\langle \psi_5 | \psi_3 \rangle} = \begin{pmatrix} S_{12} & S_{13} & S_{15} & S_{18} \\ S_{22} & S_{23} & S_{25} & S_{28} \\ S_{72} & S_{73} & S_{75} & S_{78} \\ S_{82} & S_{83} & S_{85} & S_{88} \end{pmatrix} =$$

$$S_{\langle \psi_5 | \psi_4 \rangle} = \begin{pmatrix} S_{12} & S_{14} & S_{15} & S_{17} \\ S_{22} & S_{24} & S_{25} & S_{27} \\ S_{72} & S_{74} & S_{75} & S_{77} \\ S_{82} & S_{84} & S_{85} & S_{87} \end{pmatrix} =$$

$$S_{\langle \psi_5 | \psi_5 \rangle} = \begin{pmatrix} S_{11} & S_{12} & S_{17} & S_{18} \\ S_{21} & S_{22} & S_{27} & S_{28} \\ S_{71} & S_{72} & S_{77} & S_{78} \\ S_{81} & S_{82} & S_{87} & S_{88} \end{pmatrix} =$$

$$S_{\langle \psi_5 | \psi_6 \rangle} = \begin{pmatrix} S_{13} & S_{14} & S_{15} & S_{16} \\ S_{23} & S_{24} & S_{25} & S_{26} \\ S_{73} & S_{74} & S_{75} & S_{76} \\ S_{83} & S_{84} & S_{85} & S_{86} \end{pmatrix} =$$

$$S_{\langle \psi_6 | \psi_1 \rangle} = \begin{pmatrix} S_{31} & S_{33} & S_{36} & S_{38} \\ S_{41} & S_{43} & S_{46} & S_{48} \\ S_{51} & S_{53} & S_{56} & S_{58} \\ S_{61} & S_{63} & S_{66} & S_{68} \end{pmatrix} =$$

$$S_{\langle \psi_6 | \psi_2 \rangle} = \begin{pmatrix} S_{31} & S_{34} & S_{36} & S_{37} \\ S_{41} & S_{44} & S_{46} & S_{47} \\ S_{51} & S_{54} & S_{56} & S_{57} \\ S_{61} & S_{64} & S_{66} & S_{67} \end{pmatrix} =$$

$$S_{\langle\psi_6|\psi_3\rangle} = \begin{pmatrix} S_{32} & S_{33} & S_{35} & S_{38} \\ S_{42} & S_{43} & S_{45} & S_{48} \\ S_{52} & S_{53} & S_{55} & S_{58} \\ S_{62} & S_{63} & S_{65} & S_{68} \end{pmatrix} =$$

$$S_{\langle\psi_6|\psi_4\rangle} = \begin{pmatrix} S_{32} & S_{34} & S_{35} & S_{37} \\ S_{42} & S_{44} & S_{45} & S_{47} \\ S_{52} & S_{54} & S_{55} & S_{57} \\ S_{62} & S_{64} & S_{65} & S_{67} \end{pmatrix} =$$

$$S_{\langle\psi_6|\psi_5\rangle} = \begin{pmatrix} S_{31} & S_{32} & S_{37} & S_{38} \\ S_{41} & S_{42} & S_{47} & S_{48} \\ S_{51} & S_{52} & S_{57} & S_{58} \\ S_{61} & S_{62} & S_{67} & S_{68} \end{pmatrix} =$$

$$S_{\langle\psi_6|\psi_6\rangle} = \begin{pmatrix} S_{33} & S_{34} & S_{35} & S_{36} \\ S_{43} & S_{44} & S_{45} & S_{46} \\ S_{53} & S_{54} & S_{55} & S_{56} \\ S_{63} & S_{64} & S_{65} & S_{66} \end{pmatrix} =$$

Matrix elements regarding the Hamiltonian

$$\langle\psi_I|\psi_J\rangle = \langle\phi_I|H_{IJ}|\phi_J\rangle =$$

Matrix elements of the complete Hamiltonian

$$\begin{aligned} \langle\psi_1|H|\psi_1\rangle &= \langle\phi_0|w_1Hf_1|\phi_0\rangle = -0.1447215508E+01 \\ \langle\psi_1|H|\psi_2\rangle &= \langle\phi_0|w_1Hf_2|\phi_0\rangle = -0.318808565E+00 \\ \langle\psi_1|H|\psi_3\rangle &= \langle\phi_0|w_1Hf_3|\phi_0\rangle = -0.318808565E+00 \\ \langle\psi_1|H|\psi_4\rangle &= \langle\phi_0|w_1Hf_4|\phi_0\rangle = -0.1720101851E-01 \\ \langle\psi_1|H|\psi_5\rangle &= \langle\phi_0|w_1Hf_5|\phi_0\rangle = -0.318808565E+00 \\ \langle\psi_1|H|\psi_6\rangle &= \langle\phi_0|w_1Hf_6|\phi_0\rangle = -0.318808565E+00 \end{aligned}$$

$$\begin{aligned} \langle\psi_2|H|\psi_1\rangle &= \langle\phi_0|w_2Hf_1|\phi_0\rangle = 0.318808565E+00 \\ \langle\psi_2|H|\psi_2\rangle &= \langle\phi_0|w_2Hf_2|\phi_0\rangle = -0.6331848711E+00 \\ \langle\psi_2|H|\psi_3\rangle &= \langle\phi_0|w_2Hf_3|\phi_0\rangle = 0.1478321999E+00 \\ \langle\psi_2|H|\psi_4\rangle &= \langle\phi_0|w_2Hf_4|\phi_0\rangle = -0.318808565E+00 \\ \langle\psi_2|H|\psi_5\rangle &= \langle\phi_0|w_2Hf_5|\phi_0\rangle = 0.5690144139E-02 \\ \langle\psi_2|H|\psi_6\rangle &= \langle\phi_0|w_2Hf_6|\phi_0\rangle = 0.5690144139E-02 \end{aligned}$$

$$\begin{aligned} \langle\psi_3|H|\psi_1\rangle &= \langle\phi_0|w_3Hf_1|\phi_0\rangle = 0.318808565E+00 \\ \langle\psi_3|H|\psi_2\rangle &= \langle\phi_0|w_3Hf_2|\phi_0\rangle = -0.1478321999E+00 \\ \langle\psi_3|H|\psi_3\rangle &= \langle\phi_0|w_3Hf_3|\phi_0\rangle = -0.6331848711E+00 \\ \langle\psi_3|H|\psi_4\rangle &= \langle\phi_0|w_3Hf_4|\phi_0\rangle = -0.318808565E+00 \\ \langle\psi_3|H|\psi_5\rangle &= \langle\phi_0|w_3Hf_5|\phi_0\rangle = 0.5690144139E-02 \\ \langle\psi_3|H|\psi_6\rangle &= \langle\phi_0|w_3Hf_6|\phi_0\rangle = 0.5690144139E-02 \end{aligned}$$

$$\begin{aligned} \langle\psi_4|H|\psi_1\rangle &= \langle\phi_0|w_4Hf_1|\phi_0\rangle = 0.1720101851E-01 \\ \langle\psi_4|H|\psi_2\rangle &= \langle\phi_0|w_4Hf_2|\phi_0\rangle = 0.318808565E+00 \\ \langle\psi_4|H|\psi_3\rangle &= \langle\phi_0|w_4Hf_3|\phi_0\rangle = 0.318808565E+00 \\ \langle\psi_4|H|\psi_4\rangle &= \langle\phi_0|w_4Hf_4|\phi_0\rangle = -0.1478321999E+00 \end{aligned}$$

$$\begin{aligned}\langle\psi_4|H|\psi_5\rangle &= \langle\phi_0|w_4Hf_5|\phi_0\rangle = -0.318808565E + 00 \\ \langle\psi_4|H|\psi_6\rangle &= \langle\phi_0|w_4Hf_6|\phi_0\rangle = -0.318808565E + 00\end{aligned}$$

$$\begin{aligned}\langle\psi_5|H|\psi_1\rangle &= \langle\phi_0|w_5Hf_1|\phi_0\rangle = 0.318808565E + 00 \\ \langle\psi_5|H|\psi_2\rangle &= \langle\phi_0|w_5Hf_2|\phi_0\rangle = -0.5690144139E - 02 \\ \langle\psi_5|H|\psi_3\rangle &= \langle\phi_0|w_5Hf_3|\phi_0\rangle = -0.5690144139E - 02 \\ \langle\psi_5|H|\psi_4\rangle &= \langle\phi_0|w_5Hf_4|\phi_0\rangle = 0.318808565E + 00 \\ \langle\psi_5|H|\psi_5\rangle &= \langle\phi_0|w_5Hf_5|\phi_0\rangle = -0.6331848711E + 00 \\ \langle\psi_5|H|\psi_6\rangle &= \langle\phi_0|w_5Hf_6|\phi_0\rangle = 0.6331848711E + 00\end{aligned}$$

$$\begin{aligned}\langle\psi_6|H|\psi_1\rangle &= \langle\phi_0|w_6Hf_1|\phi_0\rangle = 0.318808565E + 00 \\ \langle\psi_6|H|\psi_2\rangle &= \langle\phi_0|w_6Hf_2|\phi_0\rangle = -0.5690144139E - 02 \\ \langle\psi_6|H|\psi_3\rangle &= \langle\phi_0|w_6Hf_3|\phi_0\rangle = -0.5690144139E - 02 \\ \langle\psi_6|H|\psi_4\rangle &= \langle\phi_0|w_6Hf_4|\phi_0\rangle = 0.318808565E + 00 \\ \langle\psi_6|H|\psi_5\rangle &= \langle\phi_0|w_6Hf_5|\phi_0\rangle = -0.1478321999E + 00 \\ \langle\psi_6|H|\psi_6\rangle &= \langle\phi_0|w_6Hf_6|\phi_0\rangle = -0.6331848711E + 00\end{aligned}$$

one body interaction

$$\begin{aligned}\langle\psi_1|h|\psi_1\rangle &= \langle\phi_0|w_1hf_1|\phi_0\rangle = -0.5145583695E + 01 \\ \langle\psi_1|h|\psi_2\rangle &= \langle\phi_0|w_1hf_2|\phi_0\rangle = -0.1081503194E + 01 \\ \langle\psi_1|h|\psi_3\rangle &= \langle\phi_0|w_1hf_3|\phi_0\rangle = -0.1081503194E + 01 \\ \langle\psi_1|h|\psi_4\rangle &= \langle\phi_0|w_1hf_4|\phi_0\rangle = 0 \\ \langle\psi_1|h|\psi_5\rangle &= \langle\phi_0|w_1hf_5|\phi_0\rangle = -0.1081503194E + 01 \\ \langle\psi_1|h|\psi_6\rangle &= \langle\phi_0|w_1hf_6|\phi_0\rangle = -0.1081503194E + 01\end{aligned}$$

$$\begin{aligned}\langle\psi_2|h|\psi_1\rangle &= \langle\phi_0|w_2hf_1|\phi_0\rangle = 0.1081503194E + 01 \\ \langle\psi_2|h|\psi_2\rangle &= \langle\phi_0|w_2hf_2|\phi_0\rangle = -0.2547346525E + 01 \\ \langle\psi_2|h|\psi_3\rangle &= \langle\phi_0|w_2hf_3|\phi_0\rangle = 0.4488685850E + 01 \\ \langle\psi_2|h|\psi_4\rangle &= \langle\phi_0|w_2hf_4|\phi_0\rangle = -0.1081503194E + 01 \\ \langle\psi_2|h|\psi_5\rangle &= \langle\phi_0|w_2hf_5|\phi_0\rangle = -0.6818901517E - 02 \\ \langle\psi_2|h|\psi_6\rangle &= \langle\phi_0|w_2hf_6|\phi_0\rangle = -0.6818901517E - 02\end{aligned}$$

$$\begin{aligned}\langle\psi_3|h|\psi_1\rangle &= \langle\phi_0|w_3hf_1|\phi_0\rangle = 0.1081503194E + 01 \\ \langle\psi_3|h|\psi_2\rangle &= \langle\phi_0|w_3hf_2|\phi_0\rangle = -0.4488685850E + 01 \\ \langle\psi_3|h|\psi_3\rangle &= \langle\phi_0|w_3hf_3|\phi_0\rangle = -0.2547346525E + 01 \\ \langle\psi_3|h|\psi_4\rangle &= \langle\phi_0|w_3hf_4|\phi_0\rangle = -0.1081503194E + 01 \\ \langle\psi_3|h|\psi_5\rangle &= \langle\phi_0|w_3hf_5|\phi_0\rangle = -0.6818901517E - 02 \\ \langle\psi_3|h|\psi_6\rangle &= \langle\phi_0|w_3hf_6|\phi_0\rangle = -0.6818901517E - 02\end{aligned}$$

$$\begin{aligned}\langle\psi_4|h|\psi_1\rangle &= \langle\phi_0|w_4hf_1|\phi_0\rangle = 0 \\ \langle\psi_4|h|\psi_2\rangle &= \langle\phi_0|w_4hf_2|\phi_0\rangle = 0.1081503194E + 01 \\ \langle\psi_4|h|\psi_3\rangle &= \langle\phi_0|w_4hf_3|\phi_0\rangle = 0.1081503194E + 01 \\ \langle\psi_4|h|\psi_4\rangle &= \langle\phi_0|w_4hf_4|\phi_0\rangle = -0.5145583695E + 01 \\ \langle\psi_4|h|\psi_5\rangle &= \langle\phi_0|w_4hf_5|\phi_0\rangle = -0.1081503194E + 01 \\ \langle\psi_4|h|\psi_6\rangle &= \langle\phi_0|w_4hf_6|\phi_0\rangle = -0.1081503194E + 01\end{aligned}$$

$$\langle\psi_5|h|\psi_1\rangle = \langle\phi_0|w_5hf_1|\phi_0\rangle = 0.1081503194E + 01$$

$$\begin{aligned}
\langle \psi_5 h | \psi_2 \rangle &= \langle \phi_0 | w_5 h f_2 | \phi_0 \rangle = 0.6818901517E - 02 \\
\langle \psi_5 h | \psi_3 \rangle &= \langle \phi_0 | w_5 h f_3 | \phi_0 \rangle = 0.6818901517E - 02 \\
\langle \psi_5 h | \psi_4 \rangle &= \langle \phi_0 | w_5 h f_4 | \phi_0 \rangle = 0.1081503194E + 01 \\
\langle \psi_5 h | \psi_5 \rangle &= \langle \phi_0 | w_5 h f_5 | \phi_0 \rangle = -0.2547346525E + 01 \\
\langle \psi_5 h | \psi_6 \rangle &= \langle \phi_0 | w_5 h f_6 | \phi_0 \rangle = 0.4488685850E + 01
\end{aligned}$$

$$\begin{aligned}
\langle \psi_6 h | \psi_1 \rangle &= \langle \phi_0 | w_6 h f_1 | \phi_0 \rangle = 0.1081503194E + 01 \\
\langle \psi_6 h | \psi_2 \rangle &= \langle \phi_0 | w_6 h f_2 | \phi_0 \rangle = 0.6818901517E - 02 \\
\langle \psi_6 h | \psi_3 \rangle &= \langle \phi_0 | w_6 h f_3 | \phi_0 \rangle = 0.6818901517E - 02 \\
\langle \psi_6 h | \psi_4 \rangle &= \langle \phi_0 | w_6 h f_4 | \phi_0 \rangle = 0.1081503194E + 01 \\
\langle \psi_6 h | \psi_5 \rangle &= \langle \phi_0 | w_6 h f_5 | \phi_0 \rangle = -0.4488685850E + 01 \\
\langle \psi_6 h | \psi_6 \rangle &= \langle \phi_0 | w_6 h f_6 | \phi_0 \rangle = -0.2547346525E + 01
\end{aligned}$$

two body interactions

$$\begin{aligned}
\langle \psi_1 g | \psi_1 \rangle &= \langle \phi_0 | w_1 g f_1 | \phi_0 \rangle = 0.3698368187E + 01 \\
\langle \psi_1 g | \psi_2 \rangle &= \langle \phi_0 | w_1 g f_2 | \phi_0 \rangle = 0.7626946289E + 00 \\
\langle \psi_1 g | \psi_3 \rangle &= \langle \phi_0 | w_1 g f_3 | \phi_0 \rangle = 0.7626946289E + 00 \\
\langle \psi_1 g | \psi_4 \rangle &= \langle \phi_0 | w_1 g f_4 | \phi_0 \rangle = -0.1720101851E - 01 \\
\langle \psi_1 g | \psi_5 \rangle &= \langle \phi_0 | w_1 g f_5 | \phi_0 \rangle = 0.7626946289E + 00 \\
\langle \psi_1 g | \psi_6 \rangle &= \langle \phi_0 | w_1 g f_6 | \phi_0 \rangle = 0.7626946289E + 00
\end{aligned}$$

$$\begin{aligned}
\langle \psi_2 g | \psi_1 \rangle &= \langle \phi_0 | w_2 g f_1 | \phi_0 \rangle = -0.7626946289E + 00 \\
\langle \psi_2 g | \psi_2 \rangle &= \langle \phi_0 | w_2 g f_2 | \phi_0 \rangle = 0.1914161654E + 01 \\
\langle \psi_2 g | \psi_3 \rangle &= \langle \phi_0 | w_2 g f_3 | \phi_0 \rangle = -0.3010363851E + 00 \\
\langle \psi_2 g | \psi_4 \rangle &= \langle \phi_0 | w_2 g f_4 | \phi_0 \rangle = 0.7626946289E + 00 \\
\langle \psi_2 g | \psi_5 \rangle &= \langle \phi_0 | w_2 g f_5 | \phi_0 \rangle = 0.1250904566E - 01 \\
\langle \psi_2 g | \psi_6 \rangle &= \langle \phi_0 | w_2 g f_6 | \phi_0 \rangle = 0.1250904566E - 01
\end{aligned}$$

$$\begin{aligned}
\langle \psi_3 g | \psi_1 \rangle &= \langle \phi_0 | w_3 g f_1 | \phi_0 \rangle = -0.7626946289E + 00 \\
\langle \psi_3 g | \psi_2 \rangle &= \langle \phi_0 | w_3 g f_2 | \phi_0 \rangle = 0.3010363851E + 00 \\
\langle \psi_3 g | \psi_3 \rangle &= \langle \phi_0 | w_3 g f_3 | \phi_0 \rangle = 0.1914161654E + 01 \\
\langle \psi_3 g | \psi_4 \rangle &= \langle \phi_0 | w_3 g f_4 | \phi_0 \rangle = 0.7626946289E + 00 \\
\langle \psi_3 g | \psi_5 \rangle &= \langle \phi_0 | w_3 g f_5 | \phi_0 \rangle = 0.1250904566E - 01 \\
\langle \psi_3 g | \psi_6 \rangle &= \langle \phi_0 | w_3 g f_6 | \phi_0 \rangle = 0.1250904566E - 01
\end{aligned}$$

$$\begin{aligned}
\langle \psi_4 g | \psi_1 \rangle &= \langle \phi_0 | w_4 g f_1 | \phi_0 \rangle = 0.1720101851E - 01 \\
\langle \psi_4 g | \psi_2 \rangle &= \langle \phi_0 | w_4 g f_2 | \phi_0 \rangle = -0.7626946289E + 00 \\
\langle \psi_4 g | \psi_3 \rangle &= \langle \phi_0 | w_4 g f_3 | \phi_0 \rangle = -0.7626946289E + 00 \\
\langle \psi_4 g | \psi_4 \rangle &= \langle \phi_0 | w_4 g f_4 | \phi_0 \rangle = 0.3698368187E + 01 \\
\langle \psi_4 g | \psi_5 \rangle &= \langle \phi_0 | w_4 g f_5 | \phi_0 \rangle = 0.7626946289E + 00 \\
\langle \psi_4 g | \psi_6 \rangle &= \langle \phi_0 | w_4 g f_6 | \phi_0 \rangle = 0.7626946289E + 00
\end{aligned}$$

$$\begin{aligned}
\langle \psi_5 g | \psi_1 \rangle &= \langle \phi_0 | w_5 g f_1 | \phi_0 \rangle = -0.7626946289E + 00 \\
\langle \psi_5 g | \psi_2 \rangle &= \langle \phi_0 | w_5 g f_2 | \phi_0 \rangle = -0.1250904566E - 01 \\
\langle \psi_5 g | \psi_3 \rangle &= \langle \phi_0 | w_5 g f_3 | \phi_0 \rangle = -0.1250904566E - 01 \\
\langle \psi_5 g | \psi_4 \rangle &= \langle \phi_0 | w_5 g f_4 | \phi_0 \rangle = -0.7626946289E + 00 \\
\langle \psi_5 g | \psi_5 \rangle &= \langle \phi_0 | w_5 g f_5 | \phi_0 \rangle = 0.1914161654E + 01 \\
\langle \psi_5 g | \psi_6 \rangle &= \langle \phi_0 | w_5 g f_6 | \phi_0 \rangle = -0.3010363851E + 00
\end{aligned}$$

$$\begin{aligned}
\langle \psi_6 g | \psi_1 \rangle &= \langle \phi_0 | w_6 g f_1 | \phi_0 \rangle = -0.7626946289E + 00 \\
\langle \psi_6 g | \psi_2 \rangle &= \langle \phi_0 | w_6 g f_2 | \phi_0 \rangle = -0.1250904566E - 01 \\
\langle \psi_6 g | \psi_3 \rangle &= \langle \phi_0 | w_6 g f_3 | \phi_0 \rangle = -0.1250904566E - 01 \\
\langle \psi_6 g | \psi_4 \rangle &= \langle \phi_0 | w_6 g f_4 | \phi_0 \rangle = -0.7626946289E + 00 \\
\langle \psi_6 g | \psi_5 \rangle &= \langle \phi_0 | w_6 g f_5 | \phi_0 \rangle = 0.3010363851E + 00 \\
\langle \psi_6 g | \psi_6 \rangle &= \langle \phi_0 | w_6 g f_6 | \phi_0 \rangle = 0.1914161654E + 01
\end{aligned}$$

Recipe (It is not the more efficient path) - we are not using that.

1. Starting the Hadamard gate to prepare the ancilla register;
2. Start with $2N+1$ qubits inputs state $|\phi_i\rangle |vac\rangle |+\rangle$, distributed over the register;
3. i) The register 1 contains $|\phi_i\rangle = f_i |vac\rangle$; ii) an N qubit $|\phi_i\rangle$ ansatz; iii) Register 2 contains $|\phi_j\rangle = w_j |vac\rangle$, the vacuum contains (zero electron) state of N -qubit register and the register 3 is a 1-qubit ancilla containing $|+\rangle$, the eigenstate $+X$.
4. The inclusion of the additional phase gate P_π for calculation of the imaginary component H_{IJ} merely replaces this initial ancilla state but the $+Y$ eigenstate.
5. An N -qubit controlled -SWAP operation is then performed, with the ancilla qubit as control,

$$\frac{1}{\sqrt{2}} (|vac\rangle |vac\rangle |0\rangle + |vac\rangle |vac\rangle |1\rangle) \quad (1)$$

6. In the first set of N register, in the place of vacuum, “ $\langle \phi_I |$ ” we must to apply the U_I which contain the Jordan Wigner mapping for NO orbitals which contain the overlap terms (the annihilation operators mapped). The U_j rotations will be referent to the usual JW mapping for the creation operators.

$$\frac{1}{\sqrt{2}} (|vac\rangle |\phi_I\rangle |0\rangle + |\phi_J\rangle |vac\rangle |1\rangle) \quad (2)$$

7. We then perform a second controlled -Swap conditioned on the ancilla qubit as before, after which register 2 is again in the vacuum state and can be ignored, yielding the entangled $N+1$ qubit state

$$\frac{1}{\sqrt{2}} (|\phi_I\rangle |0\rangle + |\phi_J\rangle |1\rangle) \quad (3)$$

8. A second Hadamard gate on the ancilla is then performed and the ancilla (register 3) is measured in the Z basis.
9. As the Hadamard test, the result of the expectation value is $\langle \hat{Z}_{anc} \rangle = Re \langle \phi_I | \phi_J \rangle$. The Hamiltonian is then measured on the state of N qubits in the register 1, resulting in an expectation on value that is conditioned on the outcome of the ancilla measurement.
10. Following the example of the NOVQE, the conditional outcome may be combined shortwise with the corresponding ancilla outcome to generate the expectation value $Re \langle \phi_I | \hat{H} | \phi_J \rangle$. Repeating the circuit with ancilla prepared as the $+Y$ eigenstate, i.e., $\frac{1}{\sqrt{2}} (|0\rangle + i |1\rangle)$, yields the imaginary components of the matrix elements, $Im \langle \phi_I | \phi_J \rangle$ and $Im \langle \phi_I | \hat{H} | \phi_J \rangle$.

1 Numeric experiments

Our proposal consists in calculate the overlap between two determinants $\langle \phi_I | \phi_J \rangle$ by measuring the probability of obtaining the $|000 \dots 000\rangle$ outcome after applying sequences of Pauli operations (from w and f strings) and normalizing the results. It runs a circuit that directly evolves from one Pauli string (e. g. f_j) to another (e.g. w_i), then measures the outcome probability. While Hadamard test is a very known alternative to compute matrix elements in a quantum computer, it requires an auxiliary qubit and uses controlled operations to calculate the real or imaginary part of the expectation value of an operator relative to a quantum state. It measures the expectation value of an operator U (like a Pauli string) with respect to a given state $|\varphi\rangle$ by performing a controlled- U operation and then measuring the auxiliary qubit.

Regarding the circuit construction by sequentially applying the Pauli operators from w_i and f_j directly to main qubits. It does not use an auxiliary qubit or controlled gates which are central to the Hadamard test. Here, the Hadamard test would

require an auxiliary qubit with a Hadamard gate and a controlled application of operators U (Pauli strings) conditioned on the auxiliary qubit. This extra qubit measurement would provide the information about the expected value $\langle \phi_I | U | \phi_J \rangle$.

Concerning the measurement strategy, our quantum circuit measure all qubits and specifically looks for the $|000 \dots 000\rangle$ state. The probability of observing this state gives information about overlap between the applied Pauli strings on the initial state $|\varphi_0\rangle$. While the Hadamard test measure the auxiliary qubit, and the probability of measuring “0” and “1” on this qubit reveals information about the real or imaginary part of the overlap (or expectation value).

In the outcomes, our code computes the probability of a specific outcome, which, when combined with coefficients of f and w , yields the overlap between two prepared quantum states. Here, if we use the Hadamard test, directly compute the expectations values of a unitary operator with respect to a state by taking advantage of interference effects induced by the controlled operation.

Thus, why do not use the Hadamard test here? The HT is not necessary for this application. The code only needs to evaluate overlaps between product states generated by Pauli strings. This can be achieved with direct state preparation and measurement without needing the interference-based measurement strategy of HT. Since Pauli operators are hermitians and can be applied directly without controlled operations, there is no need for the controlled- $-U$ setup used in Hadamard test. Measuring the specific outcome probability for all qubits ($|000 \dots 000\rangle$ state) is a straightforward way to evaluate the overlaps when working with states constructed with simple Pauli strings.

In summary, for our SCGVB ansatz the quantum algorithm does not require an auxiliary qubit controlled gates. Instead, it leverages the direct Pauli strings and state probability measurements.

1.1 Circuit depth and gate complexity

As we commented our code directly applies Pauli operators (X , Y , Z) on the main qubits based on the specific strings w and f . These operations are single-qubit gates, which are fast and simple to implement on a quantum circuit. Since no auxiliary qubit or controlled operations are used, the circuit depth remains low, and each Pauli string application is efficient. If compared to Hadamard test, it requires an additional auxiliary qubit and the application of controlled operations (like controlled-Pauli gates). Controlled gates typically increase the circuit depth and induce more complexity, specifically for hardware with connectivity constraints.

Concerning to qubit resources we can straight compare the both algorithms. Our algorithm, use only the qubits needed to represent the main system state (in this case, 8 qubits). It does not require any additional qubits for measurements purposes, which conserves quantum resources. In parallel, the HT requires an auxiliary qubit in addition to the main qubits. If we compare with the modified HT presented in the Whaley’s paper, we can see the huge difference from N to $N+1$ and $2N+1$, what represent a limitation to use the quantum hardware with limited number of qubits.

1.2 Measurements

Our algorithm measure all qubits at the end of the circuit and simply checks for the probability of obtaining the all zero state $|000 \dots 000\rangle$. This straightforward measurement scheme avoids the need for post-processing on complex interference patterns. For HT, it is required to measure only the auxiliary qubit, but the interpretation of the results depends on interference patterns created by the controlled operations. Extracting the real and imaginary part of the overlap requires calculating and interpreting probabilities based on interference, which adds computational overhead.

1.3 Error propagation

Our algorithm is less sensitive to error accumulation because single-qubit gates (Pauli gates) have lower error rates on most quantum hardware. No controlled operations on entanglement are introduced which typically means fewer decoherence and noise effects.

1.4 Calculation simplicity

Our code uses a simple probability of calculation based on the counts of the $|000 \dots 000\rangle$ state, which normalized factor applied to obtain the overlap. This straightforward approach minimizes computational steps and resources. If we compare with HT algorithm, it requires computing expectation values through interference, which involves calculating both the probability of the auxiliary qubit being 0 or 1 and extracting either the real or imaginary component. This adds an extra layer of computational complexity in interpreting the measurements. This code has significantly lower complexity than HT in terms of circuit depth, qubit resources, measurement interpretations, and error resilience. For applications involving simple Pauli string overlaps, this direct approach is more efficient and better suited for near-term quantum devices. The Hadamard test, though powerful, is generally more complex and would be preferable for cases requiring expectation values of more complex operators or states where controlled operations are necessary.